Lesson 12 Teacher's Key: Hearing It Happen

Unreal Engine (optional advanced track) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new Blueprint mechanism this week - Play Sound 2D was already fully established by Lesson 3, so this lesson is purely about wiring it into the correct existing Custom Events and Branches (from Lessons 4, 6, 10, 11).

Level 1 - Full Worked Solution

- Play Sound 2D(sfx_hit.wav) joins HandleEnemyDestroyed right after the EnemiesRemaining/Score updates, before the nested Branch Lesson 11 built.
- Play Sound 2D(sfx_wave.wav) and Play Sound 2D(sfx_lose.wav) are the two genuinely new sounds, wired into the exact two Branch cases Lesson 11 and Lesson 10 already built.
- Every sound node this week is chained directly off the same Custom Event graph that already handles its corresponding state change - no separate audio Blueprint or Actor Component is needed for a game this size.

Why This Structure

- Chaining Play Sound 2D nodes directly into the existing HandleEnemyDestroyed/HandleEnemyEscaped graphs (rather than a separate audio-manager Blueprint) keeps this lesson's additions proportional to the game's actual complexity - the same restraint shown in Lessons 9 and 10 when building new UI/state systems only when genuinely needed.
- Placing sfx_win.wav and sfx_wave.wav in opposite cases of the same nested Branch (rather than an extra condition check) guarantees they can never both fire on the same event, for free, from Lesson 11's existing structure.

Level 2 / Level 3 Talking Points

- Ask students to trace the full graph for a wave-3 clear: HandleEnemyDestroyed -> nested Branch True case -> GameActive = false, Play Sound 2D(sfx_win.wav), SetStatus - a good comprehension check tying this week's addition to last week's structure.
- For students on the C++ track: point out that UGameplayStatics::PlaySound2D() is the exact same function whether called from five different spots or wrapped in a small helper - a natural discussion of when extracting a helper function is worth it versus not, for a game this size.

Common Student Mistakes

- Wiring sfx_wave.wav's Play Sound 2D node outside the False case of the nested Branch, causing it to also fire on the true win.
- Forgetting to import a raw .wav as a proper Sound Wave asset before referencing it in a Play Sound 2D node.
- Placing a sound node in HandleEnemyEscaped before the Lives -= 1 update rather than after, which is harmless here but inconsistent with every other engine's ordering this week.